

A Patient Centred Approach to Rare Disease Technology

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ABSTRACT

Rare disease patients face a long, arduous process to obtain a correct diagnosis. This often involves multiple misdiagnoses, leading to undue treatments and surgeries; patients are left feeling unheard by their healthcare providers and are even isolated from their peers who 'just don't understand'. Research to support clinicians in diagnosing rare disease patients has made great strides. However, little research aims to support the patients through this challenging time; and those that do, demonstrate minimal patient input in the design process. This is the first paper that uses methodologically robust patient driven design of pre-diagnostic technologies for rare disease patients. We conducted two workshops with rare disease patients to design a probe technology and subsequently evaluated the probe using a questionnaire. This revealed that rare disease patients predominantly wanted clinician communication aids (90.5%) and social support (61.9%) from technology, rather than the informational support that is typical with patient technology.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**; • **Applied computing** → **Consumer health**.

KEYWORDS

rare disease, patient centred design, co-design, participatory design

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1 INTRODUCTION

In this paper, we present work to explore the role of digital interventions for diagnostic support in individuals affected by rare disease. Our interest is driven by the lengthy process of diagnosis, often referred to as a diagnostic odyssey, with one in four patients

waiting five to thirty years for a diagnosis [6]. Rare diseases cover any type of disorder and are defined only by their prevalence, a disease is rare if it affects less than 200,000 people in the US [8] and less than 1 in 2,000 people in the UK [9]; so the manifestations of these disorders vary significantly and similarities in experience are only a result of their rarity.

While rare diseases are, of course, individually rare, collectively they affect approximately one in ten Americans [20], so as a group they are common. Moreover, the majority of rare disease patients were misdiagnosed at least once with 37% of patients misdiagnosed three or more times [19]. Delayed diagnoses and misdiagnoses result in delayed treatments and mistreatments, which can have detrimental effects on patients' health [6]. Consequently, our work seeks to discover opportunities within this space to support the diagnostic journeys of patients and present a design space for digital intervention on patient technology.

A number of studies have been conducted considering diagnostic support tools for rare diseases which are primarily focused on the clinician's perspective [2–4, 7, 11, 18, 21]. However, patients live with their conditions and know their symptoms intimately, in such detail that brief consultations cannot do them justice. It follows that patient involvement plays a crucial role in obtaining a diagnosis [6]. The UK Strategy for Rare Diseases [5] outlines several policies to be implemented in order to support diagnosis and early intervention. One strategy is recognising the role patients can play to help clinicians with referral and diagnosis decisions when they have access to suitable resources.

Clinician facing works in this area are well established, however research into tools for the patients to use themselves as part of the rare diagnostic journey is an emerging area; with works such as Kühnle et al.'s [17] paper on the design of RarePairs, a social media platform using peer matching for rare disease patients to aid diagnosis. While this study prioritises patient facing technology, it did not take a co-design approach to involve patients throughout the design process. This has been identified as a crucial factor in the adoption of technologies, particularly in the domain of health technologies [16]. While there are many user studies for patients with common diseases, diagnostic experiences can vary significantly from rare disease patients, so we cannot assume their needs from technology would be the same. On the contrary, their needs likely differ since the average rare disease patient consulted five clinicians, received three misdiagnoses and waited four years before receiving a diagnosis [19]. For patients to play an active role in their healthcare and to be empowered during their journey towards

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diagnosis, it is important they have access to necessary resources. Therefore, we present the first paper that uses methodologically robust patient driven design of pre-diagnostic technologies for rare diseases. Namely, rare disease patients are placed at the centre of the design process of a probe technology designed to support them by involving them at each stage to ensure this research is centered around their needs and desires. This consisted of firstly, a workshop to understand their needs during their journey towards diagnosis, secondly, another workshop to explore what technologies can better equip them and inform design choices of technology and finally, a questionnaire to evaluate what aspects of this technology are useful and explore potential for improvements. This showed rare disease patients desire technology that goes beyond typical information retrieval tasks, but instead prefer technology to support them with clinician-patient communication and finding peers who understand their experiences.

2 METHODOLOGY

A three stage study was conducted: two workshops and a probe evaluation in the form of a questionnaire. The first workshop aimed to identify common challenges, barriers and significant moments on the journey to diagnosis as well as areas that could be improved in the future. The second workshop aimed to determine how technology can support rare disease patients during their journey toward diagnosis, including which devices, information capture, insight extraction, interfaces and functionalities might be useful. This approach was taken in order to involve rare disease patients at each of the three stages of the design. Firstly, defining the problem statement - for the research to be wholly patient centred, patients need to be involved in this process since they understand best the most significant problems that they face. Secondly, exploring opportunities for technological interventions - involving patients directly in the design of technology is important to include their desired features and to avoid superfluous features [16]. And finally, evaluating the probe - after creating the probe, it is important to re-evaluate the design to establish improvements and refinements, a key principle in participatory design [10].

As part of the studies, we collected pre-collection surveys, post-collection surveys and written notes which were all collated by the lead researcher. Sessions were also recorded and transcripts were written up. These transcripts were analysed thematically according to the theoretical method described by Braun and Clarke [1] by the lead author to see general themes in the experiences of rare disease diagnosis. More specifically, open coding was used for every segment of text that related to patients' experiences of diagnosis or their attitudes towards health technologies. Once this was finished, codes were evaluated and modified where relevant. Following this, initial themes were identified, reviewed for coherence or distinctiveness and then finalised.

2.1 Participants

The two workshops were designed to have the same participants throughout each stage in order to build upon the previous stages. A total of five participants were identified via an email sent out to university staff and students in addition to an existing pool

of previously identified eligible participants and through snowballing methods. Participants were included in the study if they self-identified to meet the following criteria; they were at least 18 years old; were diagnosed with a rare disease in the UK and were able to give their informed consent. We excluded participants if their time to diagnosis was less than one year, in order to eliminate people who did not experience the hurdles typical in a rare diagnosis and as such would have less to offer in discussions. Recruitment of rare disease patients presents challenges due to their low visibility and lack of social, collective identification. This resulted in a small sample size, so restrictions on location of diagnosis were necessary due to the variability of healthcare systems. In particular, general practitioners act as gatekeepers for any secondary care in the UK's National Health Service (NHS), and NHS consultations have no direct cost. These factors can differ in other countries, such as the US [12].

For the probe evaluation, 21 participants were recruited via rare disease groups on social media. The workshop participants were also invited to participate in the evaluation. Participants were included in the study if they met the following criteria; they were at least 18 years old either with a rare disease or a Syndrome Without A Name (SWAN) and were able to give their informed consent. Restrictions on location and time to diagnosis were removed due to the increased scale of the study. Participants were also included if they had a SWAN, 'a genetic condition so rare that it is often impossible to diagnose' [25]. This would include people who have not yet received a diagnosis, making them more suited to evaluate the app.

3 WORKSHOP 1: WHAT ARE THE KEY PROBLEMS RARE DISEASE PATIENTS FACE?

The five participants for workshop one were invited to participate in a one hour workshop conducted online via Zoom. Before this study, participants provided (via a preliminary questionnaire) basic information about their journey to diagnosis in order to shape the questions of the workshop around their experiences. In particular, to establish participants' information seeking behaviour during their journey towards diagnosis, whether they interacted with patient organisations or support groups and if there were any key moments which supported their diagnosis. This informed the workshop to include more questions on information seeking behaviour due to large amounts of time spent seeking health information. There was also a focus on engagement with patient groups and organisations as, although there were high levels of engagement reported in the questionnaire, participants were only introduced to them after visiting specialists. Therefore, the workshop could explore the types of support gained from these activities as well as additional support that could be beneficial.

As a welcome to the workshop, a brief ice breaker activity was held to allow participants to become more engaged and comfortable with one another. Participants were asked to discuss experiences of diagnosis anonymously via a Mentimeter board, for it to be revealed shortly to the group, exemplifying the shared experiences among the group. The majority of the session then followed a semi-structured focus group approach to investigate participants' own

experiences of their diagnostic journey and their barriers to achieve diagnosis.

3.1 Results

The initial activity to list words which best described participants' journeys towards diagnosis revealed it was a “*difficult*” and emotionally straining time (described as “*depressing*”, “*frustrating*” and “*demoralising*”). The analysis of the questions, responses and discussions provided by the workshop's participants yielded three major themes: a perceived lack of care from healthcare professionals; difficulty explaining experiences to peers, and issues with using technology to support their journey towards diagnosis.

3.1.1 Perceived lack of care from healthcare professionals. The most predominant theme was a perceived lack of care from healthcare professionals which was expressed by all participants. Participants identified a number of issues such as; a lack of faith in their doctor's abilities; feelings of dismissal and the need to convince their doctor of their state of health; being passed on from one doctor to the next; poor social interactions with doctors; difficulties and delays with getting referrals. This theme presented an issue with the doctor-patient relationship resulting in diminished trust and communication issues. For example, one participant said “*there was no one was taking ultimate responsibility of me*”, while another stated “*no one's listening to me*”, and another “*I've been told that what was happening to me wasn't real and I ended up in A&E*”. These hindered relationships are likely due to a combination of issues faced by rare disease patients, including the high number of misdiagnoses [6, 19]; patients being disregarded and ‘accused’ of being neurotic or a hypochondriac [19] as well as significant delays and wait times for correct diagnosis [6, 19]. One of the participants also reported experiencing ‘brain fog’ with their condition, which may contribute to challenges with clinician communication, as they not remember to mention certain symptoms during consultations.

3.1.2 Difficulty explaining to peers. Another significant theme that arose from the first workshop was participants having difficulty explaining their condition to friends and family, with a number of them feeling blamed for their condition. Others lacked a support network, found peers had unrealistic expectations and were treated like they are a burden. This was likely due to non-rare individuals being unable to relate to challenges faced by rare disease patients, with one participant stating “*most of my family still do not understand*” ten years after their diagnosis.

3.1.3 Issues with technology. There were also a number of challenges with utilising existing technologies including: difficulty finding and understanding information; difficulty accessing online health information; misinformation, unreliability and a lack of trust of forums. For example, one patient said “*there are so many conditions with those [my] symptoms, so it was a bit difficult to actually come across [my condition]*”. So, patients are not satisfied with available technology; they need technology which will give them a voice and support them in a meaningful way.

3.2 Intervention Opportunities

Our analysis and feedback from participants, clearly shows rare disease patients have multiple different types of needs, including

issues surrounding information gathering needs, peer support and interactions with healthcare professionals. While many health technologies support informational needs (i.e. information seeking), rare disease patients may benefit from technologies which support them with clinical interactions as well as emotional and social needs.

4 WORKSHOP 2: HOW CAN TECHNOLOGY SUPPORT PATIENTS EXPERIENCING THESE PROBLEMS?

We invited the participants from workshop one to return for a second online workshop on Zoom that focused on low fidelity visual probes to provoke thoughts and ideas of technology that would be helpful in the rare disease diagnostic journey. We devised three visual probes based on workshop one (i.e. difficulty remembering symptoms due to brain fog, present in a number of rare diseases associated with cognitive impairment including Fabry disease [23] and ME [15]) and literature surrounding the rare patient journey (i.e. the lengthy time to diagnosis [6, 19] and difficulty understanding or remembering conversations with healthcare providers [6]). Firstly, a search engine which uses search history as a symptom list to refine future searches; secondly, a chat bot for messaging when experiencing symptoms which would gather more information on symptom manifestation and analyse symptoms for potential diagnoses; thirdly, an automatic transcription device for doctor consultations with optional additional inputted notes from the patient. These probes were designed to satisfy three main attributes: to expand participants' thinking beyond basic or existing technologies; to narrow participants' thinking from unrealistic technology and finally to leave room to encourage participants to come up with their own ideas. Participants were debriefed on the key areas for improvement identified in Workshop 1 to explore technological interventions for these. After the workshop, participants were given the opportunity to share additional thoughts.

4.1 Results

It was evident from the icebreaker activity at the start of the workshop participants had negative views on the use of technology to support them during diagnosis, with participants describing online health information as “*opinionated*” and “*confusing*”.

4.1.1 General Comments.

- Three of the five participants expressed that mobile applications would be best due to their portability
- One wanted support from technology to continue following diagnosis, with links to support groups related to their diagnosis
- Three said the app should be freely available to the public with no requirements or recommendations needed from a doctor
- Three found that existing technology for supporting diagnostic decisions was misleading or elevated their health anxiety; so they did not wish to receive patient-facing diagnostic suggestions, with one participant expressing at the start of the workshop a dislike of pre-diagnostic patient-facing technology in general

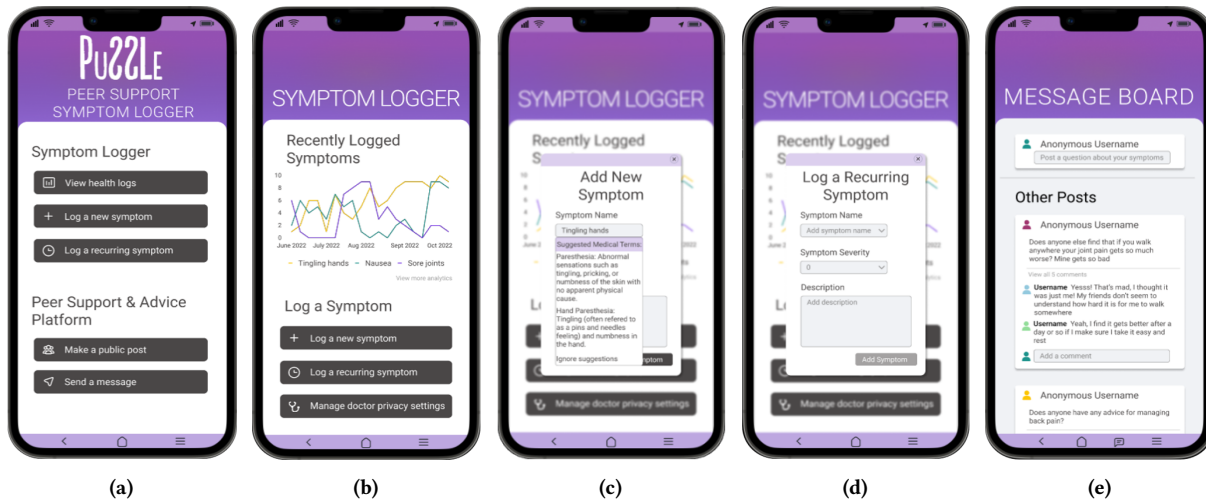


Figure 1: Probe application (Puzzle) screenshots: (a) Home screen, (b) Symptom logger main page, (c) Logging new symptoms, (d) Logging recurring symptoms, and (e) Message board from patient matched groups.

4.1.2 Clinician-Patient Mediation. The idea to use technology to provide clinician-patient mediation arose from exploring the potential of technology to alleviate the issues raised in the previous workshop, namely a perceived lack of care from healthcare professionals. Specifically, participants were interested in using a symptom logger to record and monitor their symptoms regularly on an app, which they could then share with their clinicians. Furthermore, it would help combat issues with forgetting important symptoms during the short consultation window. Upon the exploration of this idea, the participant who was previously against the use of pre-diagnostic technology expressed they had changed their mind and felt that the symptom logger ‘would be beneficial’. The high regard of this idea was likely because of the significant communication issues and perceived lack of care from one’s clinician. Additional comments for mediator functionalities include:

- Two of the five participants wanted the option to share their symptom data anonymously to databases, or non-anonymously to family members, whereas the other participants had concerns about sharing this data
- All participants wanted to be able to notify their doctor of certain symptoms using the app and felt it would save time as symptoms could be quickly assessed as benign from a quick view rather than from a time-consuming appointment and one participant felt that monitoring severity of symptoms and a reminder to log symptoms would be important
- All participants felt that home remedies and basic symptom management suggestions would be helpful

4.1.3 Peer Support. Participants expressed an interest in emotional and mental health support specific to their condition and experiences during their journey towards diagnosis. This led to the suggestion of similarity analysis to be performed on symptoms logged to connect people with their peers going through similar experiences. Profiles could be matched one-to-one or could be clustered into social networking groups to facilitate peer support and advice.

Participants expressed keen interest in this idea, but had concerns over the need for regulation and a desire to prevent over-posting. For example, they would prefer it if the app only allows group interaction after a week or so of logging to prevent people joining and interacting impulsively. Also, they were wary of hearing about other people’s experiences without a clinician’s advice. Participants felt that providing an option to interact only in peer support or only in advice or in both would help to alleviate this. In addition, it would be necessary to add a disclaimer stating to not make any medical decisions without consulting a doctor. Clearly, if this app were to be implemented, strategies must be put in place to protect users’ privacy. As the design was purely conceptual for this paper, this is not an issue at this stage. Additional comments for peer support functionalities included:

- Three of the five participants expressed interest in having other filters in addition to the peer support and advice filters and two participants wanted to be able to flag posts to improve the relevancy of posts viewed
- Most participants wanted to use both messaging and public platforms but one participant would prefer a message board to one-to-one messages

5 PROBE DESIGN AND EVALUATION

5.1 Probe Design

Directly based on the desired functionalities identified in Workshop 2 (i.e., a symptom logger, with clinician-patient mediation and peer support), a probe was designed on Figma. We named the probe Puzzle due to the puzzling nature of a rare diagnosis and this app is intended to act as a companion to patients during the challenging period of diagnosis.

Figure 1 shows the main screens of Puzzle. Puzzle consists of two main parts: a symptom logger where users can log their symptoms, the symptom severity and a description of their symptom

	Strongly Agree	Agree	Neither Agree nor Disagree	Disagree	Strongly Disagree	Total Positive	Total Negative
The app presented in the video would have been helpful during my diagnostic journey	33.3%	33.3%	9.5%	14.3%	9.5%	66.7%	23.8%
If it was available, I would have used the app presented in the video during my diagnostic journey	38.1%	33.3%	4.8%	9.5%	14.3%	71.4%	23.8%
I would have benefited from using the symptom logger to manage my health information before I was diagnosed	47.6%	33.3%	4.8%	4.8%	9.5%	81.0%	14.3%
I would want to share these symptom logs with my clinician	52.4%	38.1%	0.0%	0.0%	9.5%	90.5%	9.5%
I would benefit from being matched to groups with similar symptom patterns	28.6%	33.3%	28.6%	4.8%	4.8%	61.9%	9.5%

Figure 2: Participant responses to Likert questions in the evaluation study

occurrence (Figures 1b and 1c) and a peer support platform which enables users to interact with other patients (Figure 1e).

The symptom logger enables users to log new and reoccurring symptoms to track how their symptoms change over time. In order to further support this, users can view graphs which show the logged severity of multiple symptoms over time. To facilitate clinician-patient mediation, these symptom logs may be shared directly with users' clinicians.

The symptom logger would also translate patient terminology into clinical terminology (Figure 1c), allowing the patient to select the word and definition from a list of suggested medical terms which match their symptom. This aimed to help users research into their health in parallel with the app, but would also improve peer to peer matching by ensuring consistent terminology.

The peer support platform provides similar functionalities to typical social media platforms, namely: message boards which allow users to post to groups; private messaging and group messaging. It would, however, differ from these platforms by creating group matches using symptom logs in order to match users to other people who have similar experiences. This aims to mitigate the issue raised that peers of rare disease patients 'just don't understand'.

5.2 Evaluation Study

This probe was evaluated in a follow up questionnaire to assess whether the resulting app design would be useful for patients with rare diseases or syndromes without a name (SWAN). Participants were asked to evaluate the probe by watching a video which demonstrates the functionalities of the probe and answering the following Likert-scale questions (ranging between strongly disagree and strongly agree):

- The app presented in the video would have been helpful during my diagnostic journey.
- If it was available, I would have used the app presented in the video during my diagnostic journey.
- I would have benefited from using the symptom logger to manage my health information before I was diagnosed.
- I would want to share these symptom logs with my clinician.
- I would benefit from being matched to groups with similar symptom patterns.

These questions were then followed with open questions for constructive feedback, namely: 'What would you most like to change about this app?' and 'What functions would you most like to add to this app?'. This helped to understand why certain features were favoured over others.

5.2.1 Results. Figure 2 shows that participants generally responded positively to the app with the majority (67%) of participants reporting the app would have been helpful during their diagnostic journey and half of them stating they 'strongly agree' with the statement. Furthermore, 71.4% said they would use the app during their diagnostic journey (38.1% strongly agree, 33.3% agree).

The best regarded feature (90.5%) was the ability to share symptom logs with one's clinician. The remaining 9.5% strongly disagreed with this statement. This is likely due to communication difficulties with clinicians, as expressed by a number of participants in the further comments question. This may be seen as unhelpful if doctor-patient trust is diminished, for example, one participant stated "The app may depend too much on clinicians listening to patient feedback on diagnosis. Clinicians would need [to] buy-in on this to be useful". This poses a difficult challenge for this research area; future work should investigate if and how clinicians can be encouraged to further involve patients and give them a greater voice in their diagnosis.

The vast majority (81%) of participants stated they would have benefited from using the symptom logger during their diagnostic journey, with one participant stating "This is a superb idea. In the past [I] have tried writing symptoms down daily and endeavoured to explain these to a doctor". This implies that information management plays an important part in effective communication between patients and their clinicians.

The most contentious feature was to match patients to groups with similar symptom patterns. While the majority of participants (61.9%) still felt they would benefit from this feature, 28% gave a neutral response and 9.5% of participants said they would not benefit from this feature. This may be due to concerns for moderation, misinformation and privacy which arose when participants were asked what they would most like to change about the app with a number of participants preferring the social aspect to be opt-in/out.

6 CONCLUSION

This paper presents the first reported robust co-design study with rare disease patients to design and create technologies that can support these individuals during the diagnosis process. Previous work has largely been focused on the clinician perspective within this process [2–4, 7, 11, 18, 21]. Work that does focus on patient perspectives often draw from existing literature to create features such as patient matching in order to promote informational support (i.e. discover possible diagnoses, coping strategies, and treatments) through rare disease patient groups [17, 22].

Our approach used exploratory and co-design workshops with rare disease patients in order to create and evaluate our design probe, called Puzzle. Puzzle, builds on the concept of Kühnle et al. [17] by not only facilitating group matching, but also including a symptom logger which could be shared to the user's clinician in order to give patients a greater voice in their diagnostic journey. Future work should investigate if and how *clinicians* can be encouraged to further involve patients in their diagnosis. Additionally, in our approach, users would be matched based on symptoms logged, rather than the 53 questions required for RarePairs, reducing mental and temporal stressors on patients.

Many existing patient-facing health technologies [13, 14, 24, 26] provide informational support (i.e. researching, discovering, exploring more information about one's health), however, Puzzle focuses on supporting information management, providing emotional/peer support and support for clinician communication with the intention to be used in parallel with informational support technologies. By working with patients to understand the challenges they face, before considering how technology could mitigate these issues, we understand that the needs of rare disease patients go beyond information seeking.

This paper provides a starting point to investigate non informational support for rare disease patients during the diagnostic journey. Future work in this area should be conducted to investigate additional methods of providing non informational support and to investigate how this support would affect rare disease patients and their information seeking behaviours.

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